

Yansen Zhang

City University of Hong Kong

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EDUCATION

2022–Present; **Ph.D. Candidate in Computer Science**, Computing, City University of Hong Kong
Supervisor: Dr. [Chen Ma](#)
Research Interests: Explainable AI (recommendation); LLMs for OR; Data Valuation
2019–2022; **M.S. in Software Engineering**, Computer Science and Engineering, Sun Yat-sen University
GPA: 93.81/100. | Supervisor: Prof. [Yubao Liu](#)
Research Interests: Sequence-aware recommendation; Graph neural network
2015–2019; **B.S. in Software Engineering**, Software College, Northeastern University
GPA: 3.66/5. | Advisor: Prof. [Guibing Guo](#)
Postgraduate Candidates Exempt from Admission Exam

PROFESSIONAL EXPERIENCES

2022.04–Present; **Research on Explainable Recommendation**, City University of Hong Kong.
➤ Proposed a general data pruning framework by Shapley values in the recommendation. This work has been accepted by conference *KDD 2025*.
➤ Proposed a general framework of explainable diversity in the recommendation. This work has been accepted by conference *ECML PKDD 2025*.
➤ Presented a unified taxonomy of diversification metrics and approaches in retrieval systems from both the search and recommendation perspectives. This work has been accepted by journal *TKDE 2024*.
2019.09–2022.06; **Research on Graph-based Sequence-aware Recommendation**, Sun Yat-sen University.
➤ Focused on graph-based sequential recommendation mainly. Proposed a framework for sequential recommendation by a self-adaptive graph neural network with future contexts. This work has been accepted by conference *ICONIP 2021*.
2018.10–2019.06; **Research on Explainable Recommendation**, Northeastern University.
➤ Undergraduate Thesis: *Research and Implementation of Explainable Recommendation Based on Feature Fusion*. Achieved excellent score for the overall work.

RESEARCH INTERN EXPERIENCES

2024.06–Present; **Research on LLMs for Operations Research**, Huawei Noah's Ark Lab.
➤ Proposed a unified framework of explainable operations research within large language models. In this work, we formulated the problem of explainable operations research within LLMs, and then we designed a unified framework to solve this problem, and also developed a benchmark to verify this issue. This work has been accepted to conference *ICLR 2025*.
➤ Proposed an iterative correction framework that improves the modeling ability in optimization. This work has been submitted to conference *NeurIPS 2025*.

CONFERENCE PUBLICATIONS

1. **Yansen Zhang**, Bowei He, Xiaokun Zhang, Haolun Wu, Zexu Sun, and Chen Ma (2025). *Counterfactual Multi-player Bandits for Explainable Recommendation Diversification*. In: *ECML PKDD*. [code].
2. **Yansen Zhang**, Qingcan Kang, Wing Yin Yu, Hailei Gong, Xiaojin Fu, Xiongwei Han, Tao Zhong, and Chen Ma (2025). *Decision Information Meets Large Language Models: The Future of Explainable Operations Research*. In: *ICLR*. [code].
3. **Yansen Zhang**, Xiaokun Zhang, Ziqiang Cui, and Chen Ma (2025). *Shapley Value-driven Data Pruning for Recommender Systems*. In: *KDD*. [code].
4. Chenhao Hu, Shuhua Huang, **Yansen Zhang**, and Yubao Liu (2022). *Learning to Infer User Implicit Preference in Conversational Recommendation*. In: *SIGIR*.

5. **Yansen Zhang**, Chenhao Hu, Genan Dai, Weiyang Kong, and Yubao Liu (2021). *Self-Adaptive Graph Neural Networks for Personalized Sequential Recommendation*. In: *ICONIP*. [\[code\]](#).

JOURNAL PUBLICATIONS

1. Shuliang Wang, Jiabao Zhu, Yi Wang, Chen Ma, Xin Zhao, **Yansen Zhang**, Ziqiang Yuan, and Sijie Ruan (2025). *Hierarchical Gating Network for Cross-Domain Sequential Recommendation*. *TOIS*.
2. Haolun Wu*, **Yansen Zhang***, Chen Ma, Fuyuan Lyu, Bowei He, Fernando Diaz, and Xue Liu (2024). *Result Diversification in Search and Recommendation: A Survey*. *TKDE*. [\[code\]](#).

ONGOING PUBLICATIONS

1. **Yansen Zhang**, Qingcan Kang, Yufei Wang, Xiongwei Han, Tao Zhong, Mingxuan Yuan, and Chen Ma (2025). *SAC-Opt: Semantic Anchors for Iterative Correction in Optimization Modeling*. In: *NeurIPS*.

PATENTS

1. Yubao Liu, **Yansen Zhang** (2021). "A method for sequential recommendation model and sequential recommendation method." Chinese patent, 202111051076.3. Sun Yat-sen University.

TEACHING EXPERIENCES

Semester B 2022/23, 2023/24, 2024/25; **GE2324: The Art and Science of Data**, City University of Hong Kong.
Semester A 2022/23, 2023/24, 2024/25; **CS2334: Data Structures for DS**, City University of Hong Kong.
2019.10-2020.06; **Mathematical Analysis**, Sun Yat-sen University.

KEY SKILLS

- **Programming:** Python, C/C++, $\text{\LaTeX}$, Markdown, Matlab, SQL.
- **Frameworks/Platforms:** PyTorch, Linux, TensorFlow, Keras.
- **Software Tools:** Git, Visio, PhotoShop, MS Office.
- **Languages:** Chinese (native), English (IELTS 6.5).

SELECTED HONORS AND AWARDS

2021–2022;	Graduate Student Second-class Academic Scholarship,	Sun Yat-sen University
2019–2021;	Graduate Student First-class Academic Scholarship,	Sun Yat-sen University
2018;	Interdisciplinary Contest in Modeling (international-level),	Honorable Mention
2017–2018;	Excellent League Member Model (top 5%, departmental),	Northeastern University
2015–2017;	Excellent Students Awards (top 5%, departmental),	Northeastern University
2015–2018;	The Second Prize Scholarship,	Northeastern University
2015–2018;	National Encouragement Scholarship,	Northeastern University